Proof Debugger

A Workflow for Finding and Repairing Gaps

Engineering Math to Pure Math

May 2026

1 The Debugger Mindset

A proof is not a poem where the mood carries the meaning. It is closer to a program with types, inputs, outputs, and invariants. But unlike a program, the compiler is another human mind. Your job is to make every transition checkable.

2 Proof Submission Template

1. **The claim:** State the theorem exactly.
2. **Quantifiers:** Write all hidden quantifiers.
3. **Definitions:** List the definitions you expect to use.
4. **Strategy:** Direct, contrapositive, contradiction, cases, induction, construction, or equality by two inclusions.
5. **Proof attempt:** Write the proof.
6. **Self-debug notes:** Mark the lines you are least sure about.

3 Feedback Rubric

Issue type	Symptom	Repair
Quantifier gap	You prove something for one object but need all objects	Start with "Let x be arbitrary" or produce the needed witness
Definition gap	You use a term without unpacking it	Replace the term by its definition
Implication gap	A line does not follow from the previous line	Add a lemma or algebraic justification
Case gap	You handle one possibility but not all	List exhaustive cases before proving them
Existence gap	You say something exists without constructing or citing it	Name the object and verify its properties
Uniqueness gap	You show there is one object but not only one	Assume two candidates and prove they are equal

Notation gap	Symbols change meaning mid-proof	Declare domains and variables before using them

4 Example Debug

Claim. If n^2 is even, then n is even.

Weak attempt. Since n^2 is even, $n^2 = 2k$. Taking square roots, $n = \sqrt{2k}$, so n is even.

Diagnosis. The last step is not justified. Being the square root of twice an integer does not mean being twice an integer.

Repair by contrapositive. We prove: if n is odd, then n^2 is odd. Suppose n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Hence

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k \in \mathbb{Z}$, n^2 is odd. Therefore, by contrapositive, if n^2 is even, then n is even.

5 Line-by-Line Checklist

1. Did every variable get introduced?
2. Did every introduced variable have a domain?
3. Did the proof use the exact hypothesis?
4. Did the conclusion match the theorem exactly?
5. Did any sentence hide the real work behind “clearly,” “obviously,” or “it follows”?
6. If proving equality of sets, were both inclusions shown?
7. If proving a universal statement, was the object arbitrary?
8. If proving an existential statement, was a witness produced?
9. If proving by induction, were the base case, induction hypothesis, and induction step all explicit?

6 Feedback Form

```
Claim:

My proof:

Where I feel uncertain:

Definitions I used:

Strategy I intended:

Feedback requested:
[ ] Check correctness
```

```
[ ] Find the first broken line
[ ] Suggest a proof strategy
[ ] Make the proof more elegant
[ ] Give a hint, not a full solution
```